MATH 111 Calculus 1

# Calculating limits

Week 1 · Lecture 2 · Stewart §1.6

Dr. Salnikova Chekeres · New Uzbekistan University · Fall 2026

## While you wait

Sit with your team — the same four as in Lecture 1. Notebook open. Two volunteers as runners today? Hands up.

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"What does $\lim_{x \to a} f(x) = \infty$ mean, if $\infty$ is not a number?"

It is a sentence about behaviour: whatever bound $M$ you name, $f(x) > M$ for all $x$ close enough to $a$. We make this precise in Lecture 3.

STAND UP

"If $\lim_{x \to a} f(x)$ exists, then $f(a)$ is defined."

**Stand up if you think this is TRUE.**

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

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## Verdict

**FALSE.** $f(x) = \dfrac{x^2 - 4}{x - 2}$ has $\lim_{x \to 2} f(x) = 4$ but no value at 2. Existence of a limit and existence of a value are independent.

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| Root law | $\lim_{x\to a}\sqrt[n]{f(x)} = \sqrt[n]{\lim_{x\to a} f(x)}$ (if $n$ is even, $\lim_{x\to a} f(x) > 0$) |

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**Key idea**

If $f(x) = g(x)$ for all $x \neq a$ near $a$, then $\lim_{x \to a} f(x) = \lim_{x \to a} g(x)$.
The limit never looks at $x = a$: cancel the factors that vanish only there.

**A little history · 1694–1696**

In week 7 you will meet a famous rule for $0/0$ that carries the name of a French marquis, Guillaume de l'Hôpital. He did not discover it: he paid Johann Bernoulli 300 livres a year for lessons and for the right to publish Bernoulli's results. The rule appeared in l'Hôpital's book in 1696 — with a polite thank-you to Bernoulli in the preface.

# Question 3

$$\lim_{x \to 1} \frac{x^2 - 1}{x - 1} = \ ?$$

**A.** Substituting gives $0/0$, which is undefined, so there is no limit.

**B.** 2

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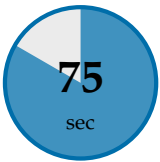
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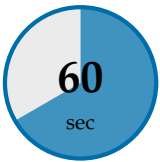
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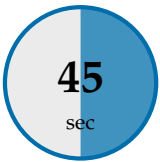
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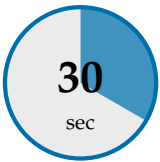
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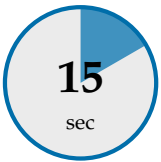
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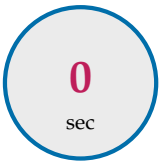
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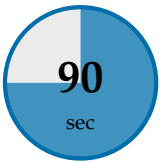
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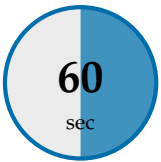

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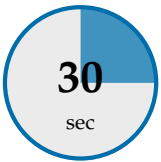

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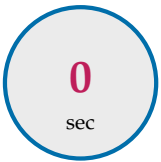

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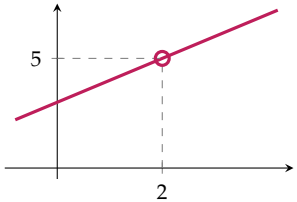
the team nearest the right aisle in row 2 — runner, please. Give the reason; I repeat what you say.

## Answer

**B.** For $x \neq 1$: $\frac{x^2-1}{x-1} = \frac{(x-1)(x+1)}{x-1} = x + 1 \to 2$. "0/0" is not a value; it is a signal that more work is needed. C repeats the Lecture-1 error: the limit does not need the value.

# Technique 1: factor and cancel

$$\lim_{x \to 2} \frac{x^2 + x - 6}{x - 2}$$

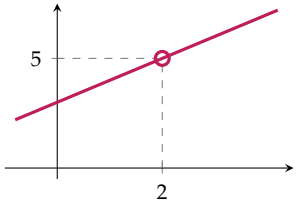



$y = \dfrac{x^2 + x - 6}{x - 2}$: the line $y = x + 3$ with a hole at $(2, 5)$.

# Technique 1: factor and cancel

$$\lim_{x \to 2} \frac{x^2 + x - 6}{x - 2}$$

$$= \lim_{x \to 2} \frac{(x-2)(x+3)}{x-2}$$



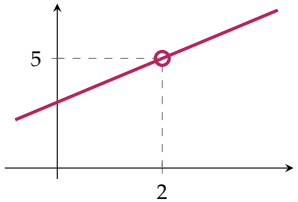

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$$= \lim_{x\to 2} (x+3) = 5.$$



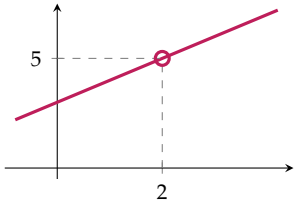

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The functions $\frac{x^2+x-6}{x-2}$ and $x+3$ agree for every $x \neq 2$; only the first has a hole at $x = 2$. The hole does not affect the limit.



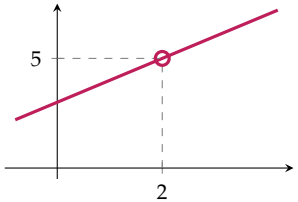

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$y = \frac{x^2+x-6}{x-2}$: the line $y = x+3$ with a hole at $(2,5)$.

**Also:**
$\lim_{h\to 0} \frac{(3+h)^2-9}{h} = \lim_{h\to 0} \frac{6h+h^2}{h} = \lim_{h\to 0}(6+h) = 6.$

$$\lim_{x \to 0} \frac{\sqrt{x+9} - 3}{x}$$

$$\lim_{x \to 0} \frac{\sqrt{x+9} - 3}{x}$$

Multiply top and bottom by $\sqrt{x+9} + 3$:

$$\lim_{x\to 0} \frac{\sqrt{x+9}-3}{x}$$

Multiply top and bottom by $\sqrt{x+9}+3$:

$$= \lim_{x\to 0} \frac{(x+9)-9}{x\,(\sqrt{x+9}+3)}$$

$$\lim_{x\to 0} \frac{\sqrt{x+9}-3}{x}$$

Multiply top and bottom by $\sqrt{x+9}+3$:

$$= \lim_{x\to 0} \frac{(x+9)-9}{x\,(\sqrt{x+9}+3)}$$
$$= \lim_{x\to 0} \frac{1}{\sqrt{x+9}+3} = \frac{1}{6}.$$

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Multiply top and bottom by $\sqrt{x+9}+3$:

$$= \lim_{x\to 0}\frac{(x+9)-9}{x\left(\sqrt{x+9}+3\right)}$$
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$$\lim_{x\to 2}\frac{\frac{1}{x}-\frac{1}{2}}{x-2}$$

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Combine the numerator: $\frac{1}{x}-\frac{1}{2}=\frac{2-x}{2x}$:

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Multiply top and bottom by $\sqrt{x+9}+3$:

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**Pattern**

Every time: rewrite until the troublesome factor cancels, *then* substitute. The rewriting is valid because $x \neq a$ throughout.

# Technique 4: absolute values and cases — go one-sided

$$\lim_{x \to 0} \frac{|x|}{x}$$

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Both sides agree, so $\lim_{x \to 1} h(x) = 2$.

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**Rule of thumb**

Whenever the formula changes at $a$ (absolute value, cases, greatest integer), compute the two one-sided limits separately and compare.

# Question 4

**Error hunt.** A student computes $\lim_{x \to 0} \dfrac{|x|}{x}$. Exactly one line is wrong. Which?

**A.** Line 1: For $x > 0$, $|x| = x$, so $|x|/x = 1$.

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**C.** Line 3: Therefore $\lim_{x \to 0^+} |x|/x = 1$ and $\lim_{x \to 0^-} |x|/x = -1$.

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THINK · silence · write your letter

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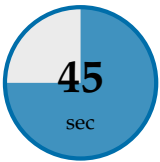
**B.** Line 2: For $x < 0$, $|x| = -x$, so $|x|/x = -1$.

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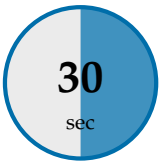
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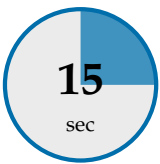
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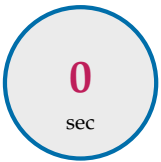
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A B C D

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**Hands down. Thank you.**

A B C D

TALK · 2 minutes

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120
sec

## In your team

Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

TALK · 2 minutes

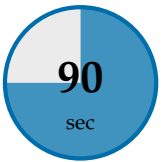
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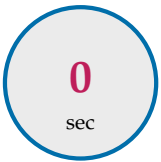
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VOTE AGAIN

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A B C D

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**Error hunt.** A student computes $\lim_{x \to 0} \dfrac{|x|}{x}$. Exactly one line is wrong. Which?

**A.** Line 1: For $x > 0$, $|x| = x$, so $|x|/x = 1$.

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**C.** Line 3: Therefore $\lim_{x \to 0^+} |x|/x = 1$ and $\lim_{x \to 0^-} |x|/x = -1$.

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**Hands down. Thank you.**

A B C D

MIC

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## Answer

**D.** Lines 1–3 are correct. When the one-sided limits differ, the two-sided limit does not exist — no averaging, no compromise: no value is approached from both sides.

STAND UP

"Since $\dfrac{x^2-4}{x-2} = x+2$, the function $\dfrac{x^2-4}{x-2}$ takes the value 4 at $x = 2$."

**Stand up if you think this is TRUE.**

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

STAND UP

"Since $\dfrac{x^2-4}{x-2} = x+2$, the function $\dfrac{x^2-4}{x-2}$ takes the value 4 at $x = 2$."

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## Verdict

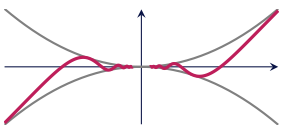
**FALSE.** The identity holds only for $x \neq 2$; at $x = 2$ the left side is $0/0$, undefined. What is true: $\lim_{x\to 2} \frac{x^2-4}{x-2} = 4$. Limit, not value.

**Theorem**

If $f(x) \leq g(x) \leq h(x)$ for all $x$ near $a$ (except possibly at $a$),

and $\lim_{x \to a} f(x) = \lim_{x \to a} h(x) = L$,

then $\lim_{x \to a} g(x) = L$.



$y = x^2 \sin(1/x)$ squeezed between $y = x^2$ and $y = -x^2$. Note: $\sin(1/x)$ itself has no limit at 0.

# The Squeeze Theorem

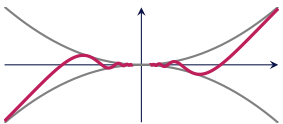
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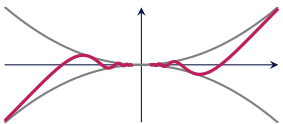
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**Example.** $\lim_{x\to 0} x^2 \sin\dfrac{1}{x} = 0.$



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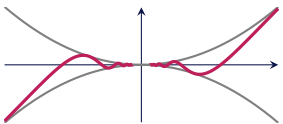
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$-1 \le \sin\frac{1}{x} \le 1$, so $-x^2 \le x^2 \sin\frac{1}{x} \le x^2$; both outer functions $\to 0$. Squeezed.



$y = x^2 \sin(1/x)$ squeezed between $y = x^2$ and $y = -x^2$. Note: $\sin(1/x)$ itself has no limit at 0.

**A little history · c. 250 BC, and Russian folklore**

Archimedes squeezed $\pi$ between the perimeters of regular 96-gons drawn inside and outside a circle: $3\frac{10}{71} < \pi < 3\frac{1}{7}$. In Russian textbooks this is the *theorem of the two policemen*: if two policemen walk to the station with a drunkard between them, the drunkard ends up at the station too, however he staggers.

# Question 5

$$\lim_{x \to 0} x \sin \frac{1}{x} = \ ?$$

**A.** $0$

**B.** $1$

**C.** Does not exist — the function oscillates.

**D.** $\infty$

THINK · silence · write your letter

$$\lim_{x \to 0} x \sin \frac{1}{x} = \ ?$$

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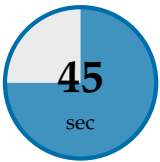
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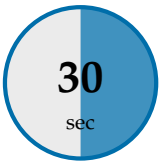
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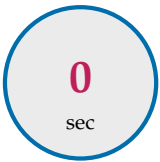
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A B C D

TALK · 2 minutes

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120 sec

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Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

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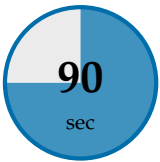
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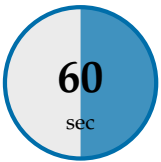
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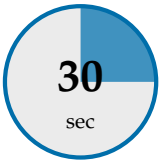
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sec

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**Hands down. Thank you.**



MIC

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the team in the middle block, row 4, next to the left aisle — runner, please. Give the reason; I repeat what you say.

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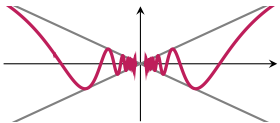
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## The microphone goes to

the team in the middle block, row 4, next to the left aisle — runner, please. Give the reason; I repeat what you say.

## Answer

**A.** $-|x| \leq x \sin \frac{1}{x} \leq |x|$ and $\pm|x| \to 0$: squeezed to 0. The oscillation (answer C) is real, but its amplitude $|x|$ shrinks to zero. Compare $\lim_{x \to 0} \sin \frac{1}{x}$, which does not exist.

## When substitution gives $0/0$ — the recipe

1. **Factor and cancel** the vanishing factor (polynomials).

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5. **Squeeze** when the function is trapped between two functions with the same limit.

**Remember**

$0/0$ is not an answer and not "undefined limit" — it is an instruction: rewrite, then substitute. The limit laws apply only when the individual limits exist and the denominator's limit is not $0$.

# EXIT TICKET

Tear a quarter page from a notebook — one per team.

1. Team name (and number, once you have one); names of those present today.
2. One sentence: *"When substitution gives* $0/0$*, I . . . "*
3. Your muddiest point — the one thing still unclear.
4. One moment today that surprised you (or made you smile).

Pass it to the aisle; runners collect. Thank you.

# BREAK
## 10 minutes

Stand up, stretch, then sit down again with your team.

Next: how do we *prove* that a limit is what we say it is? The precise definition (Lecture 3).

Course channel on Telegram:
**For Every Epsilon**
@forever_epsilon